Poster: System Architecture for Progressive Augmented Reality

Yunha Han, Chunggi Lee, Sanghoon Kim, Sungahn Ko
{diana438,cglee,seiker,sako}@unist.ac.kr
Ulsan National Institute of Science and Technology
Ulsan, Republic of Korea

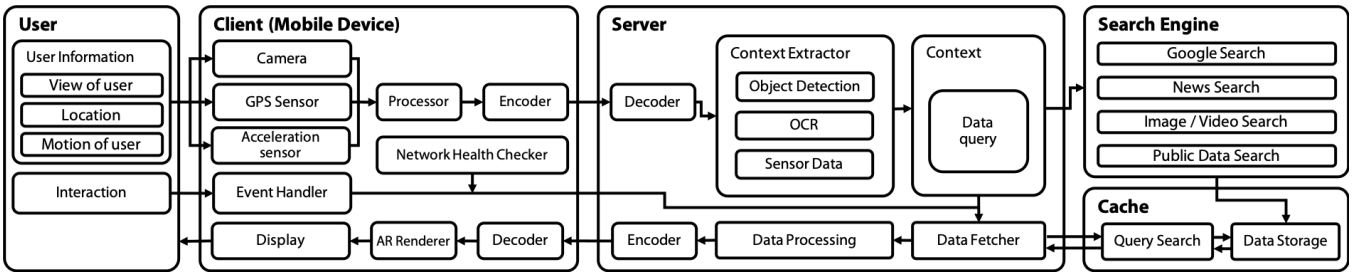

Figure 1: Progressive Augmented Reality System Architecture

CCS CONCEPTS

• **Human-centered computing → Mixed / augmented reality**; **Smartphones**; *Ubiquitous and mobile computing systems and tools.*

ACM Reference Format:
Yunha Han, Chunggi Lee, Sanghoon Kim, Sungahn Ko. 2019. Poster: System Architecture for Progressive Augmented Reality. In *The 17th Annual International Conference on Mobile Systems, Applications, and Services (MobiSys '19), June 17–21, 2019, Seoul, Republic of Korea.* ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/3307334.3328605

1 INTRODUCTION

In spite of the evolution of Augmented Reality (AR) technology, it is not wide spread in everyday life. There may be many reasons, but one of them is that it has been developed for very specific users, such as researchers and professionals. [1] To overcome this problem, Grubert et al. proposed the pervasive AR [3]. It is not limited to a specific situation, but is usable in various instances and providing continuous and flexible AR experience. AR browser is the example of utilizing pervasive AR. AR browser [4] understands the context of the user and provides corresponding information. However, if the corresponding information to the context of user cannot reach the user in time due to massive data transmission, unexpected network congestion, poor service quality or signal strength, it cannot be guaranteed to be continuous. This leads to a degradation of the user experience, and it cannot support pervasive AR.

This paper presents *Progressive Augmented Reality*, the way which quickly sends incomplete, yet informative, response about the user's current context rather than waits for sending complete information to the user. The concept of Progressive Augmented Reality comes from Progressive Data Science [6].

Our system is aware of network quality by collecting various network health parameters. According to the network status quality, it divides the chunk information to the optimal number and transmits the one that have the highest priority among the divided information. By conducting the above process iteratively, all the divided information is updated. Our client-side system utilizes Android ARCore and has been tested on Google Pixel 2XL.

2 SYSTEM ARCHITECTURE

In this study, we modularize the system architecture and construct iterative pipeline for progressive AR as shown in Fig. 1.

2.1 System Part

Our system consists of client, server, search engine, and cache.

Client: First, it sends and collects user's touch interaction and context data from the environment. Then, it checks the current network state quality. Next, it informs the server of the appropriate maximum number of objects to receive in the current network state. Lastly, it renders the object on the AR environment with received corresponding data from the server.

Server: First, it analyzes the context by receiving the context data from the context extractor. After that, it sends the context data query to the search engine and data fetcher simultaneously. The data fetcher is aware of the current network quality of the client, and determines the number of pieces of information to send. Lastly, the server provide context-oriented information to the client by the importance priority of sub-divided information.

Search Engine: It searches the various types of data with a query from the extracted context and stores data in the cache. This search result is used to provide data as feedback for the current context of the user. Data searching parses from the search results based on the web. The search engine gives the priority to the parsed data by using the search result sequentially in accordance with the pagerank algorithm[5] provided by the web search engine itself.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
MobiSys '19, June 17–21, 2019, Seoul, Republic of Korea
© 2019 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-6661-8/19/06.
https://doi.org/10.1145/3307334.3328605

Cache: It stores the retrieved data from the Search Engine. The data fetcher sends a data query to the cache, and the query search determines whether there is matched data for the query. If the data is in storage, it fetches the information with higher priority. This fetch process is done iteratively. By iteratively executing the fetch process, high priority data is given to the user first, and the detailed information is gradually provided.

2.2 Implementation Method

The client was implemented to run on android and was tested using Google Pixel 2XL. The server and search engine were implemented in Python. The object detector of the context extractor used YOLO v3 and the tessrector library was used for OCR. The search engine is based on web search, which is implemented with the Python libraries beautifulsoup and selenium. The chache is implemented in MySQL and stores data by using each data query as a table.

3 EXAMPLE SCENARIOS

Based on understanding the context of the user with OCR, object detection and sensor data, the server conducts a google search, news search, image/video search and sends the information that the user needs. In the front-end, the mobile device shows it with various data types, such as images, texts, videos, and web pages by augmenting them on the camera view which the user is watching.

We present two example scenarios that we are expecting from our system. Firstly, when a theater is detected in AR browser, it can provide a text box indicating the label of a place(or an object), a trailer video, a web page showing the show-time of movies, a ranking list of movies and URL linked to the page for discount events. The user can select or search more by interacting with the visual contents. Secondly, when a date is detected, AR browser can show weather information, stock market information of the date and titles with thumbnail images of the most viewed articles.

4 FUTURE WORK

We plan to further optimize both the back-end and front-end parts. To enhance UX, efficient view management techniques used for the optimization of labelling layout of objects rendered progressively in AR browser are required. Optimizing each modularized part of our architecture is also required for faster operation, more awareness and more detailed information. Each module can be optimized by replacing it with a better model or algorithm. We also plan to do a research more about prioritization strategies and the user's viewport direction can be an option [2].

ACKNOWLEDGMENTS

This work was partly supported by the National Research Foundation of Korea(NRF) grant funded by the Korean government(MSIT) (No. 2017R1C1B1002586). This work was supported by the Institute of Information & Communications Technology Planning & Evaluation(IITP) grant funded by the Korean government(MSIT) (2017-0-00692, Transport-aware streaming Technique Enabling Ultra Low-Latency AR/VR Services).

REFERENCES

[1] Ronald T Azuma. The most important challenge facing augmented reality. *Presence: Teleoperators and Virtual Environments*, 25(3):234–238, 2016.

[2] X. Corbillon, G. Simon, A. Devlic, and J. Chakareski. Viewport-adaptive navigable 360-degree video delivery. In *2017 IEEE International Conference on Communications*, pages 1–7, May 2017.

[3] Jens Grubert et al. Towards pervasive augmented reality: Context-awareness in augmented reality. *IEEE Transactions on Visualization and Computer Graphics*, 23 (6):1706–1724, 2017.

[4] Tobias Langlotz, Thanh Nguyen, Dieter Schmalstieg, and Raphael Grasset. Next-generation augmented reality browsers: rich, seamless, and adaptive. *Proceedings of the IEEE*, 102(2):155–169, 2014.

[5] Lawrence Page, Sergey Brin, Rajeev Motwani, and Terry Winograd. The pagerank citation ranking: Bringing order to the web. Technical report, Stanford InfoLab, 1999.

[6] Cagatay Turkay, Nicola Pezzotti, Carsten Binnig, Hendrik Strobelt, Barbara Hammer, Daniel A Keim, Jean-Daniel Fekete, Themis Palpanas, Yunhai Wang, and Florin Rusu. Progressive data science: Potential and challenges. *arXiv preprint arXiv:1812.08032*, 2018.